

Thesis Abstract

Judicial Bottleneck Diagnosis and Explainable Case Outcome Prediction in Tanzania Using SEMDR-Lite + ADAPT

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Research Area

Artificial Intelligence, Machine Learning, Natural Language Processing, Explainable AI, Digital Governance, and Data-Driven Decision Support Systems.

Problem Statement

Judicial systems in many developing countries face persistent challenges including case backlog, procedural delays, limited analytical visibility, and increasing demands for institutional accountability. Traditional approaches to judicial performance assessment often lack predictive capabilities and data-driven mechanisms for supporting evidence-based reforms. These limitations hinder efforts to improve efficiency, transparency, and access to justice.

Objectives

The study aimed to:

- Diagnose judicial bottlenecks using large-scale legal datasets.
- Develop explainable Artificial Intelligence models for legal outcome prediction.
- Evaluate cross-jurisdictional model transferability across East African legal systems.
- Explore responsible AI applications for justice sector modernization and legal reform.

Methodology

The proposed framework integrates SEMDR-Lite semantic retrieval and representation learning with ADAPT-inspired adaptive reasoning. Machine Learning and Natural Language Processing techniques were applied to judicial datasets collected from Kenya and Tanzania. Explainability was achieved through LIME and SHAP analyses to enhance transparency, interpretability, and trustworthiness of predictive outcomes.

Key Results

The proposed framework achieved an Accuracy of 0.83 and Macro-F1 score of 0.81 on the Kenyan test set, while achieving an Accuracy of 0.79, Macro-F1 score of 0.77, and ROC-AUC of 0.84 on the Tanzanian pilot dataset. The results demonstrate that AI-assisted legal analytics can support judicial performance assessment, evidence-based reform initiatives, and decision-support processes while maintaining interpretability and accountability.

Development Impact

This research contributes to Sustainable Development Goal (SDG) 16: Peace, Justice and Strong Institutions by demonstrating how Artificial Intelligence can support transparent, accountable, and data-driven justice sector reform. The proposed framework provides a foundation for responsible AI adoption within judicial institutions and broader digital governance ecosystems, supporting institutional effectiveness and evidence-based policymaking.

Keywords: Artificial Intelligence, Machine Learning, Natural Language Processing, Explainable AI, Digital Governance, Judicial Analytics, Legal Reform, SDG 16